

1. Semantic Understanding and Instruction Following (Text Version)

1. Course Content

1. Learn the basic gameplay of the large model understanding user instructions.
2. After running the large model program, the user can interact with the robot through text.
The text generation large model and vision multimodal model accurately understand the user's instructions, and finally the robot performs the specified actions according to the user's instructions and replies to the user.

2. Preparation

2.1 Content Description

The virtual machine used by default with this product starts the Dify docker environment automatically at boot, so no additional startup is needed.

2.2 Starting the Dify Environment

Open the virtual machine terminal:

```
sh ~/bringup_dify.sh
```

```
yahboom@yahboom:~$ sh ~/bringup_dify.sh
[+] up 13/13
✓ Container docker-sandbox-1      Running      0.0s
✓ Container docker-web-1          Running      0.0s
✓ Container docker-db_postgres-1  Healthy     22.5s
✓ Container docker-ssrf_proxy-1   Running      0.0s
✓ Container docker-redis-1        Running      0.0s
✓ Container docker-weaviate-1     Running      0.0s
✓ Container docker-worker-1       Running      0.0s
✓ Container docker-api_websocket-1 Running      0.0s
✓ Container docker-plugin_daemon-1 Running      0.0s
✓ Container docker-worker_beat-1  Running      0.0s
✓ Container docker-api-1          Running      0.0s
✓ Container docker-nginx-1        Running      0.0s
✓ Container docker-init_permissions-1 Exited       28.4s
yahboom@yahboom:~$
```

3. Running the Case

3.1 Starting the Program

Open the virtual machine terminal and enter the command to start the underlying proxy + camera topic:

```
ros2 launch microros_v2_bringup car_base.launch.py
```

```
yahboom@yahboom:~$ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-23-21-30-23-024975-yahboom-608172
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [608193]
[INFO] [camera_driver-2]: process started with pid [608195]
[INFO] [robot_state_publisher-3]: process started with pid [608197]
[INFO] [joint_state_publisher-4]: process started with pid [608199]
[INFO] [ekf_node-5]: process started with pid [608201]
[micro_ros_agent-1] [1784813423.243996] info      | UDPv4AgentLinux.cpp | init
                        | running...          | port: 8090
[robot_state_publisher-3] [INFO] [1784813423.323961815] [robot_state_publisher]:
got segment base_footprint
[robot_state_publisher-3] [INFO] [1784813423.324159300] [robot_state_publisher]:
got segment base_link
[robot_state_publisher-3] [INFO] [1784813423.324177934] [robot_state_publisher]:
got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1784813423.324185448] [robot_state_publisher]:
got segment cam1_Link
[robot_state_publisher-3] [INFO] [1784813423.324192058] [robot_state_publisher]:
got segment cam2_Link
[robot_state_publisher-3] [INFO] [1784813423.324198417] [robot_state_publisher]:
```

In another terminal, enter the command:

```
ros2 launch multi_brains llm_agent_control.launch.py
```

After initialization is complete, the following content will be displayed:

```
G_MODE=False, LANGUAGE=en, CAR_TYPE=ptz
[llm_agent_node-1] [INFO] [1787124487.050689131] [llm_agent_node]: Initializing
LLM Agent Node...
[llm_agent_node-1] [INFO] [1787124487.064727877] [llm_agent_node]: [CONFIG] cmd_
vel_topic=/cmd_vel
[llm_agent_node-1] [INFO] [1787124487.072992364] [llm_agent_node]: [CONFIG] imag
e_cache_path=/home/yahboom/microros_ws/cache/llm_camera.jpg
[llm_agent_node-1] [INFO] [1787124487.074496599] [llm_agent_node]: [PTZ] startup
initialization enabled: x=0.0°, y=-45.0°, count=20, interval=0.10s
[llm_agent_node-1] [INFO] [1787124487.101579243] [llm_agent_node]: [FOLLOW] scan
_topic=/scan, obstacle_guard=True, buzzer_topic=/beep, buzzer=True
[llm_agent_node-1] [INFO] [1787124487.102229731] [llm_agent_node]: [DIFY] Initia
lize client: base_url=http://localhost/v1, api_key=app-XRi6****2oQW
[llm_agent_node-1] [INFO] [1787124487.104903895] [llm_agent_node]: [DIFY] Testin
g connection....
[llm_agent_node-1] [INFO] [1787124487.118214646] [llm_agent_node]: [TRACK_KCF] S
ubscribed /tracking_bbox, type=smart_tracker/msg/TrackBox
[llm_agent_node-1] [INFO] [1787124487.118811527] [llm_agent_node]: LLM Agent Nod
e initialized successfully
[llm_agent_node-1] [INFO] [1787124487.140445756] [llm_agent_node]: [DIFY] Connec
tion successful ✓
[llm_agent_node-1] [INFO] [1787124489.075825666] [llm_agent_node]: [PTZ] Servo i
nitialization complete: x=0.0°, y=-45.0°
```

Then open another terminal:

```
ros2 run multi_brains terminal_client
```

Subsequently, enter dialogue commands in this terminal.

```

yahboom@VM:~$ ros2 run multi_brains terminal_client
[INFO] [1787124553.882304199] [terminal_client]: Terminal client initialized

> =====
>   MicroROS LLM Agent Terminal Client
> =====
> Enter natural-language commands to control the robot
>
> Examples:
>   > Go to point A
>   > Follow the red line
>   > Look at what is ahead
>   > Move forward 2 meters
>
> Commands:
>   reset - Reset conversation context
>   clear - Clear screen
>   help  - Show this help
>   exit  - Exit program
> =====
>

```

3.2 Test Cases

Reference test cases are given here. Users can create their own dialogue commands.

- Please tell me a joke about a kitten and a puppy.
- Please first move forward 1 meter quickly, then slowly move backward 0.5 meters like a turtle, then turn left 30 degrees and turn right 90 degrees.

3.2.1 Case 1

Enter the test case in the text dialogue terminal. After waiting for the model to think, the model replies to the user and executes actions according to the user's instructions.

```

> exit - Exit program
> =====

> Please tell me a joke about a kitten and a puppy.
>
> [Sending] Please tell me a joke about a kitten and a puppy.
> [Sent; you can enter another command. The latest command takes priority.]

>
> [Progress] [decision] thinking...
>
> [Robot Response]
> -----
> Here's a joke for you: A kitten and a puppy were arguing about who was smarter
> . The puppy said, 'I can learn tricks!' The kitten replied, 'That's cute. I trai
> ned my human to open doors for me, fill my bowl on command, and clean up after m
> e. Who's smarter now?'
> -----
> [Executed Actions]
> -----
> ["finishtask()"]
> -----
>
>

```

3.2.2 Case 2

The method is the same as Case 1. Enter Case 2 in the terminal, and the model will reply and execute actions according to the instructions.

```
cision plan: 1.turn left 30 degrees
[llm_agent_node-1] [INFO] [1787127899.375444816] [llm_agent_node]: [PROGRESS] [a
ction 1/1] turn_left(30, 0.5)
[llm_agent_node-1] [INFO] [1787127899.376083276] [llm_agent_node]: [MOTION] turn
left: angle=30.0°, speed=0.5rad/s
[llm_agent_node-1] [INFO] [1787127900.482957480] [llm_agent_node]: [PROGRESS] [a
ction 1/1] turn_left(30, 0.5) succeeded ✓
[llm_agent_node-1] [INFO] [1787127903.388431131] [llm_agent_node]: [PROGRESS] [f
eedback decision] Step 3 completed. Starting step 4: turning right 90 degrees
[llm_agent_node-1] [INFO] [1787127903.389261080] [llm_agent_node]: [PROGRESS] De
cision plan: 1.turn right 90 degrees
[llm_agent_node-1] [INFO] [1787127903.390600231] [llm_agent_node]: [PROGRESS] [a
ction 1/1] turn_right(90, 0.5)
[llm_agent_node-1] [INFO] [1787127903.391250769] [llm_agent_node]: [MOTION] turn
right: angle=90.0°, speed=0.5rad/s
[llm_agent_node-1] [INFO] [1787127906.608287156] [llm_agent_node]: [PROGRESS] [a
ction 1/1] turn_right(90, 0.5) succeeded ✓
[llm_agent_node-1] [INFO] [1787127909.632564588] [llm_agent_node]: [ACTION] fini
sh task: task cycle complete
[llm_agent_node-1] [INFO] [1787127909.633516096] [llm_agent_node]: [RESULT] All
steps completed successfully. Task finished.
[llm_agent_node-1] [INFO] [1787127909.634222834] [llm_agent_node]: [INPUT] == C
ommand #4 completed, feedback rounds <= 12, total time: 27.36s ==
```